

Dipeolu Ayomide Michael

Python Engineer | AI & ML Systems | Backend APIs | LLM Integration | Data Pipelines

📍 Lagos, Nigeria.

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🐙 github.com/AyDippy

🔗 dipeolu-ayomide-main-port.vercel.app

Profile

Python Engineer with a strong background in AI systems, ML engineering, and scalable backend development. Experienced in building and deploying LLM-powered applications across GCP, AWS, and Azure, with practical expertise in LLMOps, Docker-based containerization, and AI product scaling using concurrency control. Has delivered production-grade RESTful APIs, SQL data pipelines, and AI agent integrations for internal tools and automation platforms. Brings a full-stack engineering mindset from model integration and cloud infrastructure to CI/CD pipelines and cross-platform API development.

Professional Experience

2024/10 – Present
Lagos, Nigeria

AI/ML Engineer

Optimus AI Labs 🌐

- Designed and deployed an AI-powered dental phone agent using Twilio, enabling 24/7 autonomous patient service and reducing front desk workload by approximately 80%.
- Architected a hybrid RAG knowledge base using PostgreSQL and Pinecone with advanced re-ranking techniques, improving response accuracy by 70% and reducing query latency by 60%.
- Engineered multi-agent AI workflows with LangGraph and LangChain, incorporating validation layers and feedback loops to ensure reliable production performance.
- Managed the full LLM lifecycle from fine-tuning with PyTorch/TensorFlow to production deployment implementing token optimisation strategies that reduced inference costs by 25%.
- Researched and validated multiple RAG architectures (hybrid, multi-query, recursive retrieval) to determine optimal strategies per use case, improving system relevance by up to 40%.
- Developed scalable backend APIs with FastAPI/Flask and PostgreSQL/MySQL to support AI services, and automated data pipelines for web scraping and preprocessing using Pandas.
- Deployed and optimised AI solutions on Railway and Runpod within Linux environments, ensuring scalable and reliable cloud operations.

2025/09 – 2025/11
Remote

AI/ML Engineer — Steganography & Product Authentication

Rivabit 🌐

- Researched steganography techniques and applied findings to build a product authentication system for pharmaceuticals, beverages, and FMCG goods.
- Built a QR code encoder using the Steno Python library to embed unique, invisible identifiers into product packaging for originality verification.
- Developed a custom decoder using the ZXing Java library capable of authenticating encoded QR codes and distinguishing genuine products from counterfeits.
- Reviewed and synthesised multiple academic papers on steganographic encoding approaches to determine the most robust and scalable technique for production use.

2023/05 – 2023/11

Remote

Data Science Intern

Hamoye AI Labs [🔗](#)

- Developed custom web scrapers to collect structured and unstructured data from diverse online sources for downstream ML pipelines.
- Cleaned and transformed raw datasets to support accurate model training and exploratory data analysis.
- Performed EDA to surface trends and inform data-driven decision-making across multiple project domains.
- Built and evaluated classification, regression, and clustering models, optimising for performance across varied tasks including time series analysis.

Technical Skills

AI & ML Engineering

LLM Integration, LLMOps, RAG (Hybrid, Multi-query, Recursive), LangChain, LangGraph, Prompt Engineering, Multi-agent Workflows, Fine-tuning (PyTorch / TensorFlow), Vector Databases (Pinecone, ChromaDB, PGVector, FAISS), Conversational AI, Voice Agents (Twilio), Claude, GPT, Gemini, Grok, Cohere

Databases & Data Engineering

PostgreSQL, MySQL, SQLite, MongoDB, asyncpg, Pandas, NumPy, Query Optimisation, Schema Design, Data Indexing, Web Scraping, Data Pipelines

Integrations & Tooling

Google Drive API, Microsoft OneDrive, SharePoint, Google Cloud Storage, Google Speech-to-Text / TTS, Tally Webhooks, Resend, Google Sheets API, OCR

Data Science & Analysis

Exploratory Data Analysis (EDA), Classification & Regression Models, Clustering, Time Series Analysis, Sentiment Analysis, ML Model Deployment

Backend Development

Python, FastAPI, Flask, RESTful APIs, SQLAlchemy (async), Alembic, JWT & RBAC, Webhook Integration, Jinja2, Microservices Architecture, Modular System Design

Cloud & Infrastructure

Google Cloud Platform (GCP), AWS, Microsoft Azure, Docker, docker-compose, Kubernetes (basic), Railway, Runpod, GitHub Actions, CI/CD Pipelines, Linux / Bash

Security & Specialised Techniques

Steganography (Steno / ZXing), QR Code Authentication, HMAC-SHA256, Anti-enumeration Design, IP Fingerprinting, Signed URL Management

Featured Projects

FPL Padi — AI Fantasy Premier League Assistant [🔗](#)

FastAPI · LangGraph · GPT-4o-mini · Gemini · Redis · PostgreSQL · Docker · Railway

- Built a conversational AI backend for Fantasy Premier League management using FastAPI and LangGraph, enabling natural language squad recommendations and player analysis powered by GPT-4o-mini.
- Implemented a natural language to SQL pipeline using Gemini 2.5 Flash to query a live database of 500+ players across 80+ attributes including xG, form, fixture difficulty, and cost metrics.
- Engineered an image analysis feature using Gemini 2.0 Flash Vision to extract structured team data, players, costs, formation, and injuries directly from FPL screenshots.
- Automated daily web scraping pipelines with Playwright and APScheduler to keep player data and FPL news current, caching results in Redis and PostgreSQL.
- Deployed on Railway via Docker with JWT authentication, multi-session chat history, and async PostgreSQL connection pooling.

Book Recommendation Assistant — Semantic Search Chatbot [🔗](#)

Flask · Scrapy · Cohere · Pinecone · MySQL

- Scraped Amazon book data using Scrapy, storing structured records in MySQL and exporting to JSON for downstream processing.

- Generated vector embeddings from book metadata using Cohere's Embed API and indexed them in Pinecone for efficient semantic retrieval.
- Built a Flask-based chatbot interface (Bookie) that embeds user queries at runtime and performs semantic search against Pinecone to surface relevant book recommendations via Cohere's chat API.
- Designed a modular project structure separating scraping, embedding, database, and web app layers for maintainability and extensibility.

Education

2019 – 2024 Abeokuta, Nigeria	Bachelor of Engineering (B.Eng.) in Electrical and Electronics Engineering (First Class) <i>Federal University of Agriculture, Abeokuta.</i>
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